

## Employee Attrition Prediction using Machine Learning

To better understand our workforce dynamics and proactively address employee turnover, we conducted a comprehensive analysis of historical HR data. The objective was to identify the underlying patterns of why employees leave and to build a predictive tool that flags "flight-risk" personnel before they hand in their resignation.

### Key Drivers of Attrition: Why Are They Leaving?

Our analysis revealed that salary alone does not dictate an employee's decision to leave. Instead, turnover is driven by a combination of compensation, burnout, and career stage. The top three predictors of attrition are:

- **Monthly Income:** Unsurprisingly, base compensation is the strongest anchor for retention. Lower-paid employees are leaving at significantly higher rates.
- **Chronic Overtime:** Employees who consistently work overtime are at a severe risk of burnout and subsequent exit, regardless of their pay bracket.
- **Age & Tenure:** Younger employees, particularly those in their first two years with the company, represent the highest flight risk. Retention stabilizes significantly as employees mature within the organization.

### High-Risk Areas: Where Should We Focus?

The data clearly points to specific areas of the business that require immediate HR intervention:

- **The Sales Department:** This division is experiencing the highest overall attrition rate (over 20%).
- **Job Roles:** Specifically, **Sales Representatives** (nearly 40% exit rate) and **Laboratory Technicians** (roughly 24% exit rate) are turning over at alarming speeds compared to the rest of the company.

### Strategic HR Recommendations

Based on these findings, we recommend implementing the following targeted retention strategies:

1. **Mandatory Overtime Audits:** Implement a strict policy requiring secondary managerial approval for sustained overtime, specifically targeting Sales Representatives and Laboratory Technicians. We must structurally prevent burnout before it results in turnover.
2. **Targeted "Stay Interviews":** Launch a proactive outreach program for younger employees approaching their 1-to-2-year anniversary. Engaging these employees during their highest flight-risk window allows HR to address concerns regarding compensation, environment, and career progression before they start interviewing elsewhere.

### System Limitations

While this predictive analysis provides a powerful early-warning system for HR, it is important to note its limitations. The system relies entirely on historical, quantifiable data. It cannot predict qualitative, deeply human factors—such as a sudden personal family emergency, an unrecorded conflict with a new manager, or an unexpected career shift. Therefore, this tool should be used to guide proactive HR conversations, not to make definitive assumptions about an employee's loyalty.